

Beyond Forecast Accuracy: Feasibility-Constrained Demand Planning

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Abstract

Operational demand planning increasingly relies on predictive models, but operations execute planning signals rather than forecast accuracy metrics. A numerically accurate forecast can still produce volatile inventory targets, frequent switching, or execution violations when downstream systems cannot absorb abrupt plan changes. This paper formulates forecast-driven demand planning as feasibility-constrained model selection. Candidate forecasts are converted into executable planning signals and evaluated using WAPE, inventory exposure, planning volatility, switching burden, execution violations, normalized planning loss, greedy-to-DP gap, and oracle gaps. DataCo execution records provide data-informed execution-risk scenarios, Favorita retail demand data provide the main forecast-to-plan proof of concept, M5 retail data test hierarchy and intermittent-demand robustness, and Walmart weekly sales data test business-context robustness. Across 123 paper-facing evaluation groups, the accuracy-first deployable strategy differs from the operational-loss-optimal deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. Favorita results show ranking shifts under severe execution-risk scenarios, M5 results show that planning grain and intermittency change the preferred deployable mechanism, and Walmart results show that business context and cadence constraints affect deployable rankings. The contribution is not a new forecasting model. It is an interpretable operations decision layer showing that forecast accuracy should be treated as a diagnostic metric, while deployment should be evaluated by feasibility-adjusted planning utility.

Keywords

Demand planning, Forecast-to-plan conversion, Operational feasibility, Planning stability, Execution risk.

1 Introduction

Demand planning is an operational deployment problem, not only a forecasting problem. Forecast-driven decisions influence inventory targets, replenishment orders, warehouse labor, transportation commitments, supplier coordination, and capacity planning. A model rollout that improves WAPE can still create abrupt changes in plans, new approval requirements, or execution workloads that the operating system cannot absorb.

Operations teams execute planning signals, not forecast error metrics. A planner cannot execute WAPE or RMSE; the organization executes an inventory target, a staffing plan, a shipment commitment, or a replenishment quantity. When the planning signal changes faster than supplier, warehouse, transportation, or enterprise-system workflows can adapt, the forecast layer and execution layer become misaligned. This paper calls that misalignment the planning-execution gap.

Recent work shows rapid progress in AI-enabled supply-chain planning and forecasting benchmarks (Walter et al. 2025; Culot et al. 2024; Vlachos and Reddy 2025; Makridakis et al. 2022). These advances make deployment evaluation more important. More accurate forecasts can be valuable, but forecast accuracy alone does not show whether the resulting plans are stable, feasible, and governable.

This paper studies feasibility-constrained model selection for operational demand planning. Candidate forecasts and planning policies are converted into executable planning signals, evaluated using inventory exposure and feasibility terms, and ranked through deployable decision-layer strategies. The empirical contribution is not only that forecast accuracy is incomplete, but that the planning-execution gap can be measured and shown to change deployable strategy rankings.

The study uses four public-data roles. DataCo supply-chain execution records anchor execution-risk scenarios from observed late-delivery heterogeneity. Favorita daily retail demand data provide the main forecast-to-plan proof of concept. M5 retail data test hierarchy and intermittent-demand robustness. Walmart weekly sales data test business-context robustness through history-only versus history-plus-context forecasts, holiday and markdown windows, weekly cadence constraints, and switch-budget sensitivity. Rossmann is reserved for future work because completed Rossmann experiment outputs were not found in the audit.

The paper makes five IEOM-oriented contributions:

1. Define planning-execution gap rate as a measurable deployment-feasibility metric.
2. Propose a forecast-to-plan feasibility layer that separates WAPE diagnostics from operational planning utility.
3. Develop a decision layer with ensemble, smoothing, greedy, DP, and budgeted-DP strategy families.
4. Use DataCo execution records to construct execution-risk stress scenarios.
5. Validate ranking shifts using Favorita, M5, and Walmart, then translate the findings into planning recommendations.

1.1 Objectives

The specific objectives of this study are:

1. Define a forecast-to-plan feasibility layer that measures operational feasibility in addition to WAPE.
2. Measure the planning-execution gap using execution violation rate and planning-execution gap rate.
3. Construct DataCo-informed execution-risk scenarios for deployment stress testing.
4. Compare accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP deployable planning strategies.
5. Validate ranking shifts across Favorita, M5, and Walmart, and translate the findings into practical planning recommendations.

2 Literature Review

2.1 AI-enabled Demand Planning and Forecasting Benchmarks

Recent supply-chain research shows increasing interest in AI-enabled demand planning. Walter et al. (2025) review artificial intelligence in demand planning and identify AI tools, supply-chain applications, and digital supply-chain impacts as major knowledge clusters. Culot et al. (2024) review empirical AI studies in supply-chain management and emphasize the need to connect AI adoption to operational research directions. Vlachos and Reddy (2025) similarly review machine learning in supply-chain management and identify planning, execution, and control as areas where machine learning can support decision making.

Forecasting benchmarks and time-series foundation models show the same trend from a forecasting perspective. The M5 competition demonstrates the scale and hierarchy of modern retail forecasting benchmarks (Makridakis et al. 2022). Recent time-series foundation models, including decoder-only forecasting models and pretrained probabilistic time-series models, show that forecasting pipelines are moving toward broader zero-shot and cross-domain capability (Das et al. 2024; Ansari et al. 2024). These studies motivate the forecasting layer of the present paper, but they do not remove the need to evaluate whether forecast-driven planning signals are operationally executable.

2.2 Forecast-to-decision and Predict-then-optimize Research

Forecast-to-decision research is directly relevant because it explains why prediction accuracy and downstream decision quality can differ. Elmachetoub and Grigas (2022) show that the standard predict-then-optimize workflow can be improved by training predictions toward decision loss rather than prediction loss alone. Mandi et al. (2024) survey decision-focused learning and describe how machine learning and constrained optimization can be integrated to improve decision quality under uncertainty. Jia et al. (2025) propose a scenario predict-then-optimize method for data-driven online inventory routing, showing that downstream routing and replenishment decisions require more than point forecast accuracy.

The present paper is related to this stream, but its purpose is different. It does not introduce a new end-to-end learning algorithm. It proposes an interpretable forecast-to-plan decision layer for operational demand planning, where existing candidate forecasts are evaluated by inventory exposure, planning volatility, switching burden, and execution-capacity violations.

2.3 Inventory Implications and Planning Stability

Inventory theory provides the operational foundation for holding and shortage costs. Silver et al. (1998) and Axsater (2015) describe inventory planning as a cost and service-level problem rather than a pure prediction problem. Forecast accuracy metrics remain important, but Hyndman and Koehler (2006) show that forecast-error measures have known limitations and must be interpreted carefully across series and contexts. Forecast combination is also relevant because weighted or averaged forecasts can reduce dependence on a single model (Bates and Granger 1969).

Modern forecasting candidate sets often include nonlinear machine-learning methods such as random forests, XGBoost, and LightGBM (Breiman 2001; Chen and Guestrin 2016; Ke et al. 2017). These models can improve forecast accuracy, but model deployment also creates governance, monitoring, and switching burden. From an operations perspective, Forrester (1961) shows that industrial systems can amplify instability when material and information flows respond with delays. Advanced planning systems similarly separate forecasting, planning, coordination, and execution layers (Stadtler et al. 2015). This separation motivates explicit measurement of planning volatility and execution-capacity violations.

Rolling-horizon decision theory also supports finite-horizon selection logic because a locally attractive action can be undesirable once future transitions are considered (Sethi and Sorger 1991). This perspective is consistent with the paper's use of greedy-to-DP comparisons to measure myopic planning suboptimality.

2.4 Research Gap

Existing studies improve forecasts, combine forecasts, or optimize downstream decisions. Fewer studies provide a practical decision layer that compares forecasting-driven strategies by inventory exposure, planning volatility, switching burden, and execution-capacity violations. This paper addresses that gap by defining planning-execution gap rate and feasibility-adjusted normalized planning loss. The goal is not to replace forecasting metrics, but to position WAPE as a diagnostic measure alongside operational feasibility.

3 Methods

The proposed method is an industrial decision framework for selecting deployable planning strategies. Figure 1 shows the four layers. Candidate forecasts and policies are first translated into planning signals. The same feasibility scoring layer then evaluates every candidate before the decision layer ranks deployable strategies. This design follows forecast-to-decision research showing that downstream operational quality can differ from forecast-error quality (Elmachetoub and Grigas 2022; Mandi et al. 2024).

Figure 1 separates diagnostic forecast accuracy from deployable planning utility. It also links method components to

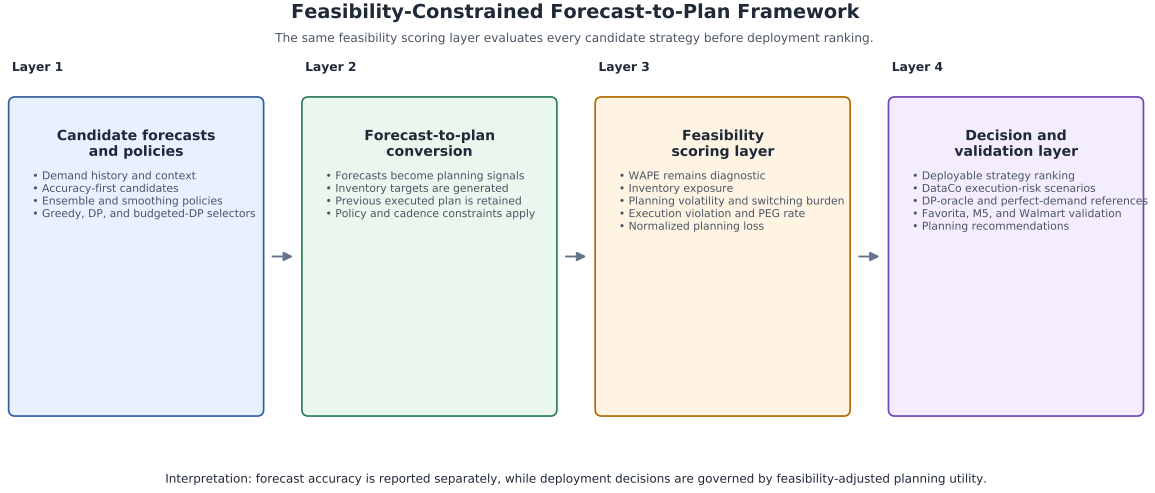


Figure 1: Four-layer framework for feasibility-constrained demand planning.

validation evidence: DataCo stress scenarios test execution pressure, DP and oracle benchmarks test decision-layer value, M5 tests hierarchy and intermittency, and Walmart tests business context and weekly planning cadence.

3.1 Forecast-to-plan Feasibility Model

Let i index a planning unit, t index a planning period, and m index a candidate forecasting model or planning policy. The candidate forecast is $\hat{y}_{i,t}^m$, and the executable planning signal is $p_{i,t}^m$. The forecast-to-plan conversion is:

$$p_{i,t}^m = g(\hat{y}_{i,t}^m, \text{policy}_t). \quad (1)$$

In this study, the planning signal is an inventory target implied by the candidate forecast and planning policy.

Inventory exposure is measured using holding and shortage terms:

$$IC_{i,t}^m = h[p_{i,t}^m - y_{i,t}]^+ + b[y_{i,t} - p_{i,t}^m]^+, \quad (2)$$

where h is holding cost, b is shortage cost, $y_{i,t}$ is realized demand, and $[x]^+ = \max(x, 0)$.

Planning volatility and execution violation are:

$$PV_{i,t}^m = |p_{i,t}^m - p_{i,t-1}|, \quad (3)$$

$$EV_{i,t}^m = [|p_{i,t}^m - p_{i,t-1}| - C_{i,t}]^+, \quad (4)$$

where $C_{i,t}$ is the execution capacity available to absorb plan movement. The planning-execution gap rate for strategy s is:

$$PEG(s) = \frac{1}{T} \sum_{t=1}^T \mathbb{I}\{EV_t(s) > 0\}. \quad (5)$$

The normalized planning loss is:

$$L(s) = \beta_{\text{inventory}} \cdot \text{NIC}(s) + \lambda_{\text{volatility}} \cdot \text{NVC}(s) + \lambda_{\text{execution}} \cdot \text{NEC}(s) + \lambda_{\text{switch}} \cdot \text{NSC}(s), \quad (6)$$

where NIC, NVC, NEC, and NSC are normalized inventory, volatility, execution, and switching components. WAPE is reported separately as diagnostic forecast accuracy. The main operational objective excludes direct forecast-error terms because realized inventory exposure and feasibility terms capture the deployable consequences of forecast-driven planning decisions.

3.2 Stepwise Feasibility-constrained Selection Algorithm

Algorithm 1. Feasibility-constrained selection procedure.

1. Generate candidate forecasts or planning policies for each planning unit and period.
2. Convert each candidate forecast into a planning signal.
3. Compute inventory exposure relative to realized demand.
4. Compute planning volatility and switching burden relative to the previous executed plan.
5. Compute execution violation under the available execution capacity.
6. Aggregate the normalized planning loss.
7. Select the deployable planning action using one of the decision policies.
8. Validate the selected strategy against scenario and oracle benchmarks.

Steps 1–3 generate candidate planning signals. Steps 4–6 compute inventory exposure, planning volatility, execution violation, and planning-execution gap rate. Step 7 produces deployable strategy rankings. Step 8 validates those rankings using DataCo scenarios, DP-oracle comparison, M5 hierarchy and intermittency robustness, and Walmart business-context and cadence robustness. The procedure is modular: accuracy-first baselines, ensembles, smoothing rules, greedy selectors, DP selectors, and budgeted-DP selectors all pass through the same feasibility layer.

Table 1 links the framework components to the evidence used in this paper and to their operational use.

Table 1: Framework components, evidence, and operational use.

Framework component	Purpose	Evidence	Operational use
Forecast-to-plan and PEG rate	Convert forecasts into executable planning signals and measure infeasible movement.	91.1% mismatch rate; 15.4% vs. 2.7% PEG-rate contrast.	Monitor feasibility alongside WAPE.
DataCo execution-risk scenarios	Stress-test strategy rankings as execution pressure changes.	Favorita baseline-to-severe ranking shifts.	Run scenario tests before rollout.
Greedy and DP selection	Compare one-period selection with finite-horizon planning.	Greedy-to-DP gap and DP-oracle gap.	Use DP when switching coordination matters.
Smoothing and ensemble policies	Test gradual adaptation and low-disruption forecast composition.	Favorita ensemble tradeoffs and M5 smoothing robustness.	Match policy to execution capacity.
Oracle, M5, and Walmart validation	Separate deployable decisions from diagnostic benchmarks and test robustness.	M5 hierarchy/intermittency and Walmart business-context outputs.	Avoid one-size-fits-all deployment rules.

3.3 Horizon-aware and Constrained Selection

The greedy selector minimizes the current-period operational cost. Let $c_t(m_{t-1}, m)$ denote the validation-derived stage cost of choosing candidate m after previous model m_{t-1} . The greedy selector chooses:

$$m_t^G = \arg \min_{m \in \mathcal{M}} c_t(m_{t-1}, m). \quad (7)$$

The finite-horizon DP selector accounts for future transition effects:

$$V_t(m_{t-1}) = \min_{m \in \mathcal{M}} \{c_t(m_{t-1}, m) + V_{t+1}(m)\}, \quad (8)$$

with terminal condition $V_{T+1}(\cdot) = 0$. The greedy-to-DP gap measures myopic selection suboptimality. When this gap is small, one-step selection may be adequate; when switching approval and coordination costs are material, DP provides a planning-ahead benchmark.

The budgeted DP selector adds a governance constraint:

$$\sum_{t=2}^T \mathbb{I}\{m_t \neq m_{t-1}\} \leq K. \quad (9)$$

The state includes the previous model and the number of switches already used. Transitions exceeding the switch budget are discarded. If missing candidate availability leaves no feasible path, the implementation uses a flagged incumbent-stays fallback that is budget-compliant when the incumbent model remains available.

Smoothing represents gradual operational adaptation:

$$p_t = \alpha p_t^{\text{candidate}} + (1 - \alpha) p_{t-1}. \quad (10)$$

Lower α values reduce execution burden but can delay response to demand changes. Ensemble policies combine candidate forecasts and are evaluated as low-disruption alternatives when accuracy remains comparable.

3.4 Validation Benchmarks

The Realized-Inventory Oracle DP is non-deployable. It uses realized test-period inventory outcomes over the same forecast candidates and measures the value of unavailable inventory-outcome information. The DP-oracle gap measures selection suboptimality relative to this non-deployable benchmark.

The Perfect-Demand Oracle is also non-deployable. It uses realized demand as the planning signal and is a diagnostic reference, not a guaranteed lower bound. Exact demand tracking can increase planning volatility and execution violations when the operation cannot absorb rapid plan movement. The perfect-demand reference gap therefore measures a diagnostic total gap that includes both forecast uncertainty and feasibility effects.

4 Data Collection and Experimental Design

The empirical design uses DataCo, Favorita, M5, and Walmart for distinct operational roles. Table 2 summarizes these roles. DataCo is not treated as an inventory-control dataset; it is used to transfer empirical heterogeneity in execution risk into scenario design. Favorita is the main forecast-to-plan proof-of-concept dataset. M5 is the hierarchy and intermittent-demand robustness dataset. Walmart is the business-context robustness dataset.

DataCo execution-risk anchor. DataCo contains supply-chain fulfillment records with late-delivery outcomes. The analysis extracts late-delivery anchors from context-level percentiles. The current generated anchors range from 0.535 at the low-risk percentile to 0.901 at the severe-risk percentile. Under the additive clipped scenario mapping, these anchors produce execution weights from 0.318 to 0.500. These are sensitivity weights, not exact dollar costs for Favorita, M5, or Walmart.

Favorita proof of concept. Favorita is a daily retail demand-planning panel from Corporacion Favorita. The pipeline treats each store-family pair as a planning unit and uses promotion, holiday, oil-price, transaction, and store-context information where available at planning time. Candidate forecasts include transparent baselines and machine-learning candidates such as LightGBM and XGBoost when dependencies are available (Ke et al. 2017; Chen and Guestrin 2016). The purpose is to test whether forecast-to-plan conversion changes deployable strategy ranking.

M5 robustness design. M5 contains 3,049 items across 10 stores, and the pipeline samples item-store series for

Table 2: Dataset roles and experimental purpose.

Dataset	Role	Data used	Purpose
DataCo	Execution-risk anchor	Late-delivery outcomes by operational context	Defines data-informed execution-risk scenarios; not treated as an inventory-control dataset.
Favorita	Main proof of concept	Daily store-family retail demand panel	Tests forecast-to-plan conversion and deployable strategy ranking under execution-risk scenarios.
M5	Robustness analysis	Hierarchical and intermittent retail demand	Tests whether the planning-execution gap persists across planning grain and demand sparsity.
Walmart	Business-context robustness	Weekly store-department sales with holiday and markdown context	Tests context-aware forecasting, holiday and markdown windows, weekly cadence constraints, and switch-budget sensitivity.

quick and medium robustness runs (Makridakis et al. 2022). The IEOM submission uses M5 to evaluate item-store, department-store, and category-store planning grains and to test intermittent-demand buckets. This design checks whether the planning-execution gap persists when demand is sparse or aggregated.

Walmart business-context robustness design. Walmart provides weekly store-department sales with holiday, markdown, store, and economic context fields. The IEOM submission uses the generated Walmart outputs to compare history-only and history-plus-context candidate forecasts, holiday and markdown stress windows, weekly cadence constraints, and switch-budget sensitivity. This design checks whether the feasibility layer remains informative when the planning problem is weekly and context driven rather than daily and store-family driven.

Rossmann is not treated as a completed experiment in this IEOM draft. It remains an optional future robustness panel.

5 Results and Discussion

5.1 Numerical Results

Finding 1. Accuracy-first and deployable-best are different decisions. Baseline validation asks whether planners need to care about feasibility. Across 123 paper-facing evaluation groups, the WAPE-best deployable strategy differs from the operational-loss-best deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. These verified values show that the planning-execution gap is measurable, not only conceptual. Table 3 summarizes the core numerical evidence.

Finding 2. Ensemble strategies can be low-disruption deployment options. Favorita quick-mode results provide the main proof of concept. In the baseline scenario, Global Best has WAPE 0.109 and normalized planning loss 1.550, while Operational Ensemble has WAPE 0.119 and normalized planning loss 1.422. Under the severe DataCo-informed execution-risk scenario, Global Best remains accurate but has normalized planning loss 2.000. Simple Ensemble and Operational Ensemble have lower normalized losses of 1.473 and 1.496, respectively. This pattern does not imply a universal winner. It shows that ensembles can be low-disruption options when execution pressure makes plan movement costly.

Finding 3. DP quantifies the cost of myopic selection. Greedy selection minimizes current-period operational cost, while DP minimizes finite-horizon operational cost. The mean greedy-to-DP gap is 0.044, and the mean DP-oracle gap is 0.198. These values are not large enough to claim that DP is always required. They support a more practical interpretation: one-step selection may be adequate in low-governance environments, while DP becomes more valuable when switching approval, coordination, or infrastructure-update costs are material.

Finding 4. Smoothing represents the stability boundary. Smoothing changes the executed planning signal rather than the underlying forecast. It can reduce execution violations by adapting gradually, but excessive smoothing can

Table 3: Core numerical evidence for the planning-execution gap.

Metric	Verified value	Interpretation
Accuracy-first mismatch rate	91.1% of 123 groups	WAPE-best deployable strategy often differs from operational-loss-best deployable strategy.
Mean accuracy-first loss gap	11.169	Average normalized-loss disadvantage of accuracy-first selection when it is not operationally best.
Accuracy-first PEG rate	15.4%	Fraction of periods with positive execution violation for accuracy-first strategies.
Best deployable PEG rate	2.7%	Same metric for the best deployable operational-loss strategy.
Greedy-to-DP gap	0.044	Measures myopic selection cost where greedy and DP are both available.
Mean DP-oracle gap	0.198	Selection suboptimality relative to non-deployable realized-inventory DP information.
DataCo execution-risk anchors	0.535 to 0.901	Late-delivery anchors used for low to severe execution-risk scenarios.

increase inventory exposure or delayed adaptation. M5 item-store results illustrate this boundary: Smoothed Alpha 0.25 is the best deployable method at the item-store grain, with WAPE 0.368, PEG rate 0.010, and normalized planning loss 1.223. This supports smoothing when execution capacity constraints dominate fast adaptation.

Finding 5. M5 shows that planning aggregation changes deployment policy. M5 reveals a critical operational principle: planning aggregation changes strategy optimality. Table 4 shows that Operational Ensemble is strongest at department-store and category-store grains, while Smoothed Alpha 0.25 is strongest at the item-store grain and across low-, medium-, and high-intermittency buckets. Low-level item-store planning is sparse and volatile, so smoothing can dampen noise. Department and category aggregation stabilize demand, allowing ensemble methods to perform better. This means one global deployment rule should not be applied uniformly across the supply-chain hierarchy.

Table 4: M5 hierarchy and intermittency robustness summary.

Robustness slice	Best deployable method	WAPE	PEG rate	Normalized loss
Hierarchy: category-store	Operational Ensemble	0.159	0.063	1.261
Hierarchy: department-store	Operational Ensemble	0.162	0.100	1.177
Hierarchy: item-store	Smoothed Alpha 0.25	0.368	0.010	1.223
Intermittency: high	Smoothed Alpha 0.25	0.616	0.030	1.284
Intermittency: low	Smoothed Alpha 0.25	0.254	0.000	1.156
Intermittency: medium	Smoothed Alpha 0.25	0.367	0.001	1.168

Finding 6. Walmart validates business-context and weekly-cadence robustness. Walmart serves a different validation role from M5. M5 tests hierarchy and intermittency, while Walmart tests whether business context and weekly planning cadence alter deployable strategy ranking. Walmart robustness contributes 20 evaluation groups, with an 80.0% accuracy-first mismatch rate in the main Walmart robustness summary. Table 5 summarizes the compact Walmart evidence. The results show that context-aware features can reduce execution burden, holiday windows can make budgeted DP attractive, and switch-budget sensitivity controls an accuracy-feasibility tradeoff rather than giving a monotonic win.

5.2 Graphical Results

Figure 2 plots the Favorita accuracy versus execution-penalty frontier. The figure shows that WAPE and execution penalty do not move together. Global Best is near the best WAPE region, but other deployable strategies produce much lower execution penalties. The operational implication is that the deployable frontier is not necessarily the WAPE-best point.

Figure 3 shows normalized planning loss across DataCo-informed execution scenarios. As execution-risk weight

Table 5: Walmart business-context robustness summary.

Validation slice	Data-driven result	Interpretation
Main Walmart robustness	20 evaluation groups; accuracy-first mismatch rate is 80.0%.	Weekly business-context planning still shows frequent separation between WAPE-best and deployable-best selection.
Context-aware features	History-plus-context reduces Global Best execution penalty by 23,745 and violation rate by 0.003 in the baseline comparison.	Business context can improve feasibility, but it does not remove the need for a decision layer.
Holiday windows	Budgeted DP is best deployable in holiday windows, with WAPE 0.068–0.071 and execution violation rate 0.005.	Switch governance is especially relevant during operationally sensitive weeks.
Weekly cadence constraint	Global Best remains best deployable under the history-plus-context weekly-cadence setting, with normalized loss 1.200.	Accuracy-first selection can remain reasonable when cadence constraints and context are favorable.
Switch-budget sensitivity	Increasing the switch budget improves WAPE from 0.059 at $K = 0$ to 0.057 at $K = 8$, while normalized loss rises from 1.206 to 1.229.	Switch budget controls an accuracy-feasibility tradeoff rather than giving a monotonic win.

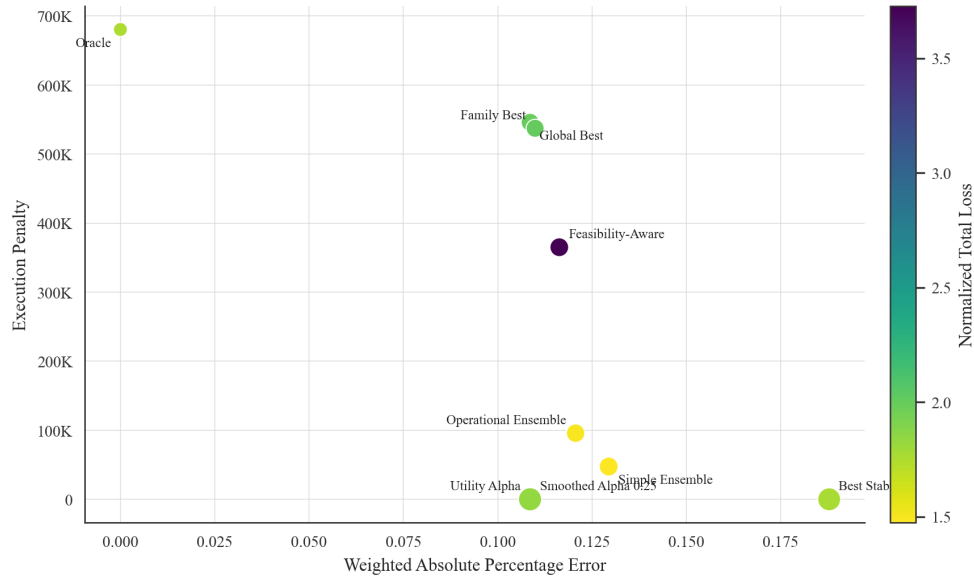


Figure 2: Favorita accuracy and execution penalty frontier.

changes, strategy ranking changes. This supports scenario testing as a deployment practice: the model selected for a flexible execution environment may not be the model selected for a constrained execution environment.

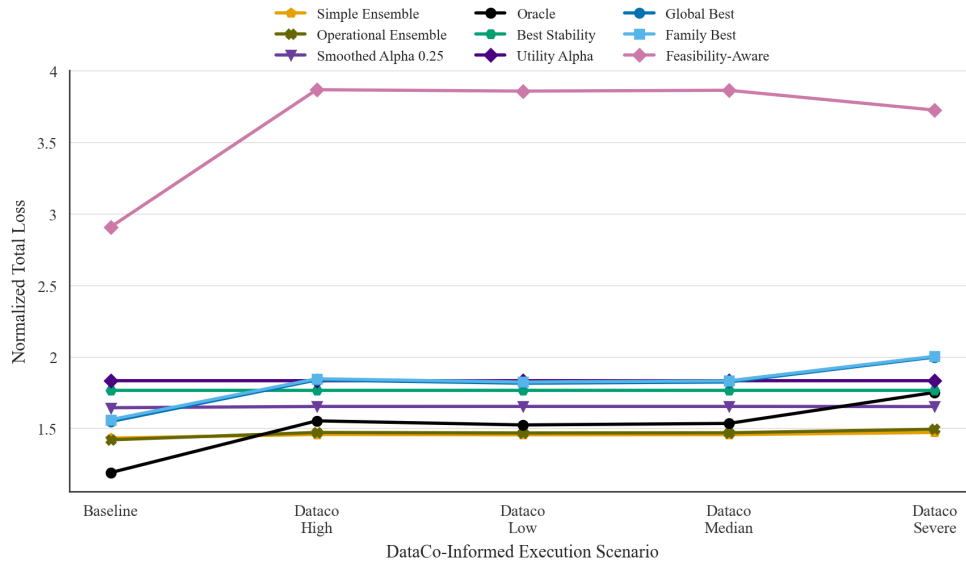


Figure 3: Favorita normalized planning loss by DataCo-informed execution scenario.

5.3 Proposed Improvements

The empirical results translate into deployment practices rather than a single universal algorithm. Table 6 summarizes the recommended use of the framework under common operating conditions.

Table 6: Deployment guidance from empirical findings.

Operating condition	Deployment risk	Recommended planning practice
Accuracy-only approval process	WAPE-best strategy may not be deployable-best.	Monitor PEG rate and normalized planning loss alongside WAPE.
Moderate execution constraint	Forecast changes may create avoidable plan movement.	Use execution-risk scenario testing before rollout.
Severe execution constraint	Fast adaptation may exceed operational capacity.	Consider smoothing when stability dominates speed.
High switching governance	Frequent model changes require approval, monitoring, or system updates.	Use DP or budgeted DP to account for switching burden.
Multi-level planning hierarchy	One deployment rule may not fit item, department, and category grains.	Choose strategy by planning grain and intermittency profile.
Holiday, markdown, or weekly cadence context	Business context can change feasibility rankings.	Run context-aware and cadence-aware stress tests before deployment.

Monitor planning-execution gap rate. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. PEG rate should therefore be monitored alongside WAPE.

Avoid WAPE-only model approval. The accuracy-first deployable method differs from the operational-loss-optimal deployable method in 91.1% of paper-facing evaluation groups. Forecast accuracy remains important, but it should not be the sole deployment criterion.

Use execution-risk and business-context stress tests. Favorita rankings change under DataCo-informed execution pressure, and Walmart shows that context, holiday, markdown, cadence, and switch-budget settings affect deployability. Model rollout should therefore include feasibility stress tests before approval.

5.4 Validation

5.4.1 Baseline Validation: Do Planners Need to Care About Feasibility?

Yes. The mismatch rate and PEG-rate contrast show that WAPE-best selection and deployable-best selection answer different questions. The study does not claim that feasibility-aware strategies always dominate; it shows that the operational deployment criterion is measurably different from the forecast-accuracy criterion.

5.4.2 Scenario Validation: Does Execution Pressure Change Strategy Ranking?

Yes. DataCo-informed scenarios validate whether execution-risk weights alter deployable ranking. The Favorita baseline-to-severe shift shows that a strategy attractive under lower execution pressure can become less attractive when execution-capacity violations are weighted more heavily.

5.4.3 Algorithmic Validation: What Does Planning Ahead Buy?

The greedy-to-DP gap measures the cost of one-period selection. The DP-oracle gap measures selection suboptimality relative to a non-deployable realized-inventory DP benchmark over the same forecast candidates. These gaps show how much value comes from horizon-aware selection and how much value remains unavailable without realized inventory-outcome information.

The perfect-demand reference gap serves a different purpose. It is diagnostic rather than a lower-bound oracle because exact demand tracking may create planning volatility that violates execution capacity. It therefore measures a total reference gap that includes forecast uncertainty and feasibility effects.

5.4.4 Robustness Validation: Does the Finding Generalize Beyond One Retail Dataset?

Yes, with context-specific mechanisms. Favorita provides the main forecast-to-plan proof of concept. M5 shows that hierarchy and intermittency change the preferred deployable method. Walmart shows that weekly cadence and business context also change deployable strategy comparisons. Together, these results support the thesis that feasibility-constrained planning is a tradeoff surface, not a single-method leaderboard.

6 Conclusion

This paper reframes operational demand planning as feasibility-constrained model selection. The planning-execution gap was quantified by comparing accuracy-first model selection with operational-loss-optimal deployable planning selection. Across 123 paper-facing evaluation groups, the accuracy-first deployable method differs from the operational-loss-best deployable method in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, while the best deployable operational-loss strategy has a gap rate of 2.7%.

Objective 1 was addressed by defining a forecast-to-plan evaluation layer that reports WAPE separately from inventory exposure, volatility, switching burden, and execution violations. Objective 2 was addressed by defining planning-execution gap rate. Objective 3 was addressed by using DataCo execution records to construct data-informed execution-risk scenarios. Objective 4 was addressed by comparing accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP deployable strategies. Objective 5 was addressed through Favorita, M5, and Walmart validation and through evidence-linked planning recommendations.

The results show that execution-risk scenarios can change deployable strategy ranking, that hierarchy and intermittency can change the preferred mechanism, and that weekly business context can affect feasibility tradeoffs. The practical implication is direct: planners should evaluate deployment feasibility before model rollout. Forecast accuracy remains important, but it should be interpreted as a diagnostic metric rather than as the sole deployment objective.

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Biography

Author biography placeholder. Replace this paragraph with a biography for each author before submission. IEOM guidance limits each biography to 250 words.

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